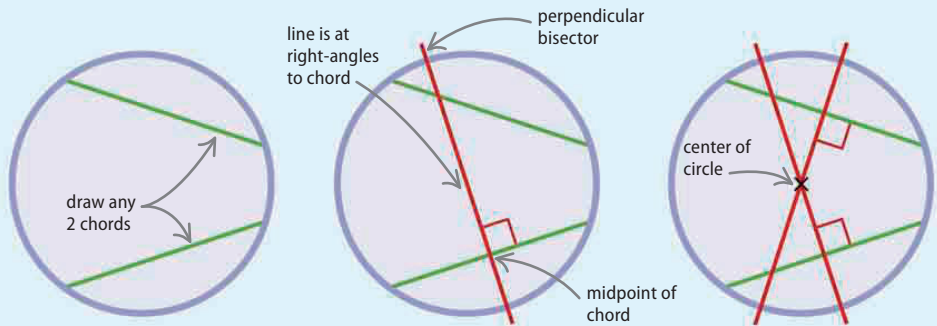


Finding the center of a circle

Chords can be used to find the center of a circle. To do this, draw any two chords across the circle. Then find the midpoint of each chord, and draw a line through it that is at right angles to that chord (this is a perpendicular bisector). The center of the circle is where these two lines cross.



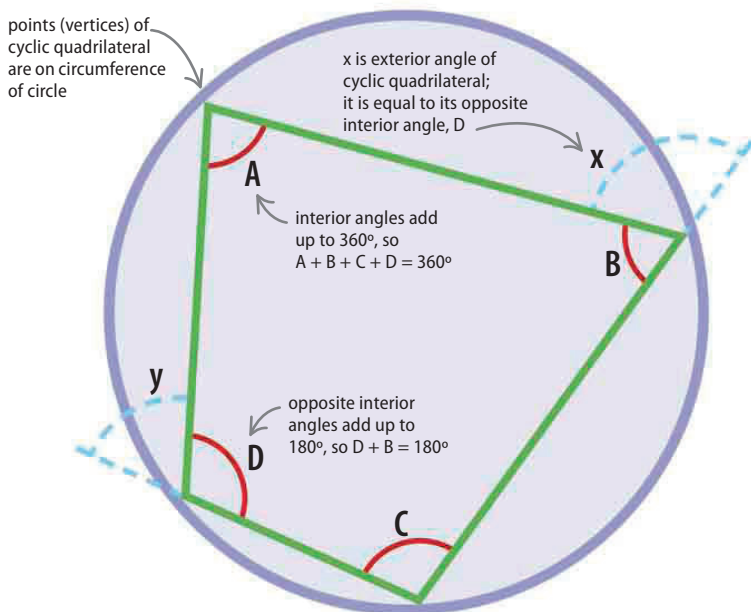
First, draw any two chords across the circle of which the center needs to be found.

Then measure the midpoint of one of the chords, and draw a line through the midpoint at right-angles (90°) to the chord.

Do the same for the other chord. The center of the circle is the point where the two perpendicular lines cross.

Cyclic quadrilaterals

Cyclic quadrilaterals are four-sided shapes made from chords. Each corner of the shape sits on the circumference of a circle. The interior angles of a cyclic quadrilateral add up to 360° , as they do for all quadrilaterals. The opposite interior angles of a cyclic quadrilateral add up to 180° , and their exterior angles are equal to the opposite interior angles.



△ Angles in a cyclic quadrilateral

The four interior angles of this cyclic quadrilateral are A, B, C, and D. Two of the four exterior angles are x and y.

$$A + B + C + D = 360^\circ$$

△ Interior angle sum

The interior angles of a cyclic quadrilateral always add up to 360° . Therefore, in this example $A + B + C + D = 360^\circ$.

$$A + C = 180^\circ$$

$$B + D = 180^\circ$$

△ Opposite angles

Opposite angles in a cyclic quadrilateral always add up to 180° . In this example, $A + C = 180^\circ$ and $B + D = 180^\circ$.

exterior angle opposite to angle B

$$y = B$$

exterior angle opposite to angle D

$$x = D$$

△ Exterior angles

Exterior angles in cyclic quadrilaterals are equal to the opposite interior angles. Therefore, in this example, $y = B$ and $x = D$.